Abdul Hakam

Electronic and Telecommunication Engineering Undergraduate

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Profile

Electronic and Telecommunication Engineering undergraduate focused on software development, artificial intelligence, machine learning, computer vision, and data-driven systems, with experience integrating software with embedded hardware. Developed full-stack business platforms, a real-time computer-vision recognition system, and embedded applications involving firmware, sensors, communication, and companion software.

Education

B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering **2024–Present**
University of Moratuwa, Sri Lanka *CGPA: 3.47/4.00 after four semesters*

GCE A/L Examination: Physical Science **2022 (2023)**
St. Anthony's College, Kandy *3A with a Z-score of 2.1072*

GCE O/L Examination **2019**
St. Anthony's College, Kandy *Results: 9A*

Experience

Owner and Operations Manager **2023–Present**
Gatronix Store *Sri Lanka*

- Own and manage Gatronix Store, a retail and wholesale computer-products business.
- Oversee purchasing, inventory, supplier and customer relationships, accounts, and day-to-day operations; also designed, developed, and now operate dedicated retail and wholesale business systems.

Technical Skills and Tools

- **Programming Languages:** C, C++, Python, Java, and JavaScript.
- **Software Development:** React, Spring Boot, Supabase, Flutter, HTML, and CSS.
- **AI, Machine Learning, and Computer Vision:** MediaPipe, OpenCV, scikit-learn, and SVM classification.
- **Embedded Systems and Interfaces:** STM32, ESP32, Arduino, Raspberry Pi.
- **Electronics and PCB Development:** Analog and digital circuit design, schematic capture, PCB layout, prototyping, soldering, and hardware debugging.
- **Development and Testing:** STM32CubeIDE, Arduino IDE, Git, and Postman.
- **Simulation and CAD Tools:** Altium Designer, Proteus, LTspice, and SolidWorks.
- **Spoken Languages:** English, Tamil, and Sinhala.

Projects

Software, AI, and Data Projects

Real-Time ASL Fingerspelling Recognizer **2026**
Python, MediaPipe, OpenCV, scikit-learn

- Developed a real-time webcam system that collects normalized 3D hand landmarks and recognizes static ASL fingerspelling using a calibrated SVM, confidence filtering, temporal voting, rejection-pose training, and CPU inference optimizations.

Gatronix Retail and Wholesale Business Systems: Retail POS | Wholesale POS **2026–Present**
React, Vite, Supabase, JavaScript

- Developed separate systems for Gatronix Store's retail and wholesale operations, with workflows tailored to sales, purchasing, inventory, payments, and customer and supplier account management.
- Built retail workflows for POS billing, stock, quotations, returns, COD and online orders, cash flow, and documents; added wholesale workflows for bulk sales, supplier orders, goods receiving, stock in transit, landed cost, credit, cheques, and reporting.

University Lost-and-Found Management System **2026**
React, Spring Boot, JWT

- Contributed to a full-stack campus platform for reporting, tracking, and recovering lost and found items through a centralized user and administrative system.
- Worked on frontend routing, authentication and user-related screens, backend integration, software testing, and supporting system documentation within the project team.

Embedded Software and Hardware Integration Projects

Servie - Autonomous Restaurant Food-Delivery Robot **2026**
ESP32-S3, STM32, C/C++, custom electronics

- Designed and integrated an autonomous restaurant delivery robot using magnetic-line navigation, encoder feedback, ultrasonic obstacle detection, tray load sensing, and motor-control electronics.
- Developed and tested the ESP32-S3 and STM32 control integration, including inter-microcontroller communication, navigation logic, battery management, and system-level hardware debugging.

FallTrack Smart Fall-Detection Watch **2025**
ESP8266, MPU6050, MAX30102, Flutter, custom PCB

- Co-developed a wearable safety device that detects falls using motion and orientation data, monitors heart rate, and provides real-time feedback through an OLED display and local alerts.
- Integrated embedded firmware, Wi-Fi communication, a Flutter companion application, custom PCB development, battery-powered operation, and a compact wearable enclosure.

WaveSync - RF-Linked Alarm Clock **2026**
Digital logic, R–2R DAC, comparators, 433 MHz OOK, Altium Designer

- Co-developed a microcontroller-free alarm system that compares digitally encoded clock data with a user-set alarm and broadcasts the resulting trigger to synchronized remote receivers.
- Designed the 433 MHz OOK transmitter and receiver circuits and contributed to schematic development, PCB layout, hardware integration, and prototype testing.

Analog Automatic Gain Controller **2026**
NE5532, JFET voltage-controlled resistor, Proteus, analog signal processing

- Designed and simulated an analog audio automatic-gain-control circuit using op-amp buffering and amplification, peak detection, feedback control, and a JFET voltage-controlled resistor.
- Built and tested the hardware prototype with function-generator and real-audio inputs, improving output-level stability while reducing loading, clipping, and distortion.

Competitions and Achievements

Robotics Design and Competition Module **2025**
Designed and built the best-performing robot for an arena-based navigation and task-completion challenge.

Sri Lankan Robotics Challenge (SLRC) **2025**
Participated in the university category of the national robotics competition.

SPARK 25/26 **2025–26**
Reached the semifinal stage with a flood early-warning system.

SLIoT Challenge **2026**
Reached the semifinal stage with a flood early-warning system.

OCTWAVE 3.0 – Kaggle Challenge 02 **2026**
Participated in the Tom & Jerry image-classification challenge, advanced to Round 2, and placed 16th.

ROSCO **2026**
Participated in the robotics competition.

Certifications

- **Data Science in Python** – Coursera, 2021.
- **Full-Stack Web Development Bootcamp** – Udemy, 2021.

Referees

Dr. Upeka Premaratne
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